



Image Denoising:

From Foundations to State-of-the-Art

Tracing how the image prior moved out of human hands — from transform-domain filtering, through convolutional networks, to attention and state-space models.

ANCHOR PAPER

DnCNN · Zhang et al.
IEEE TIP 26(7), 2017

SOURCES

15 total · 12 Category A
78.6% / 21.4%

SPAN

2007 → 2024
Four eras · three bottlenecks

AGENDA

What this deck covers

Structured to the five sections required by the Stage 1 guidelines, in order.

I

PART I — THE PROBLEM

- 04 Topic definition & the core problem
- 05 Why denoising is mathematically hard

II

PART II — THE EVOLUTION

- 07 The visual evolutionary timeline
- 08 The three bottlenecks

III

PART III — THE SOURCES

- 10 Master list — Category A
- 11 Category B & the 70/30 arithmetic
- 12 Category B justification

IV

PART IV — PAPER BY PAPER

- 14–24 A1–A12, one slide each
- 25 How the papers connect
- 26 The direction we propose

PART I

The problem

One equation with two unknowns per pixel. Every method in this review answers a single question — where does the prior come from?



Recovering a clean image from a corrupted observation

THE TOPIC

Image denoising. Selected because the literature treats it as foundational, not merely representative. SCUNet calls it perhaps the most fundamental image restoration problem: a testbed for evaluating image priors and optimisation algorithms, pluggable into larger restoration pipelines, and underpinning downstream work broadly.

WHY IT STILL MATTERS

Medical necessity. ZS-N2N motivates its work by noting that reconstructing fine detail is especially important for medical images, where fine details are necessary for accurate diagnosis.

Still unsolved. SCUNet, published 2023, states that existing methods mostly rely on simple noise assumptions and that a general-purpose blind denoising method for real images remains unsolved.

THE DEGRADATION MODEL — FROM THE ANCHOR PAPER

$$\mathbf{y} = \mathbf{x} + \mathbf{v}$$

Recover clean image $\mathbf{x}$ from noisy observation $\mathbf{y}$. The common assumption is that $\mathbf{v}$ is additive white Gaussian noise with standard deviation σ .

— *DnCNN*, §1

This is the sentence that keeps the project from being a history lesson: the problem is open, not closed.

Three difficulties, each stated by a paper in our list

1

The problem is ill-posed

One observation, two unknowns per pixel. Infinitely many (x, v) pairs explain any y . Restormer states it generally: because of its ill-posed nature it is a highly challenging problem that usually requires strong image priors for effective restoration.

2

A prior must be supplied

DnCNN frames this from a Bayesian viewpoint: when the likelihood is known, image prior modelling plays the central role — non-local self-similarity, sparse, gradient and Markov random field models. The classical shape is a hand-designed regulariser, re-solved per image at test time.

3

Smoothing destroys detail

Noise removal and detail preservation pull in opposite directions. BM3D names the tension in its own claim: collaborative filtering reveals even the finest details shared by grouped blocks while at the same time preserving the essential unique features of each individual block.

THE QUESTION THAT ORGANISES EVERYTHING THAT FOLLOWS

If a prior is unavoidable, then the history of denoising is the history of where the prior comes from — and who or what writes it.

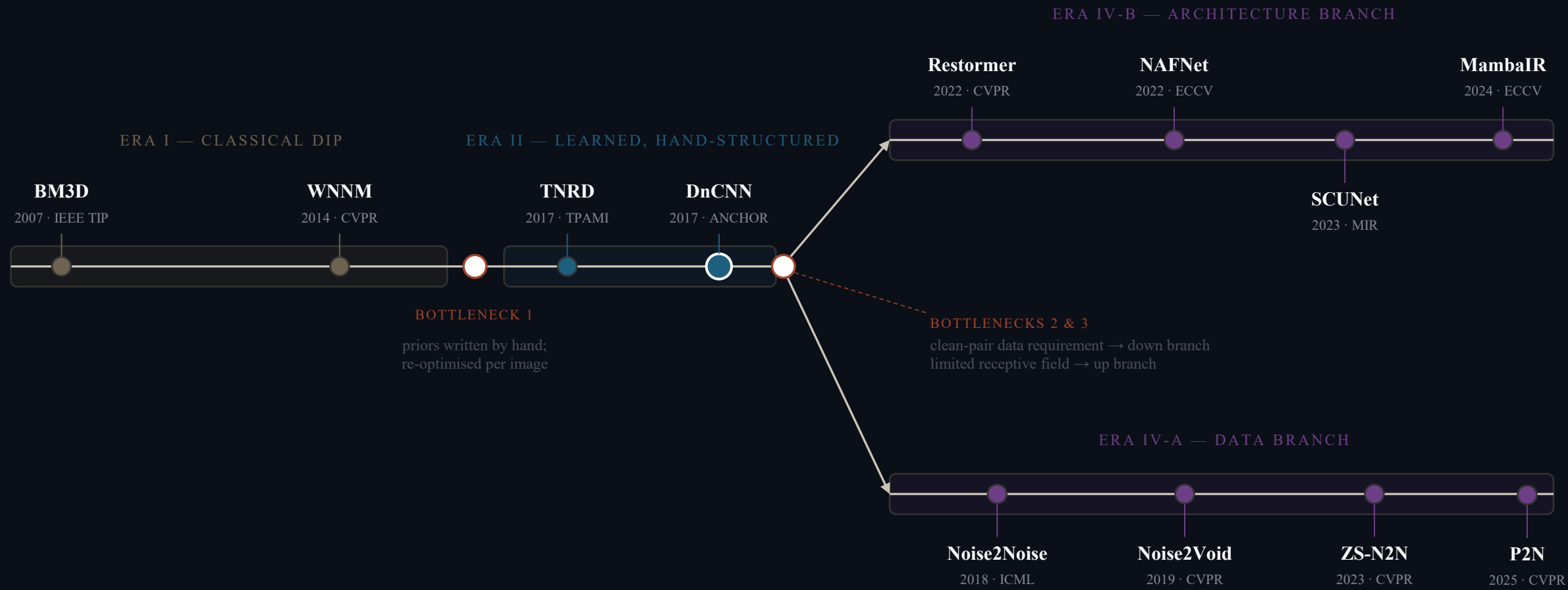
PART II

The evolution

Four eras, separated by three bottlenecks. Each bottleneck is stated by the paper that broke it — so the arc is evidenced from primary sources, not imposed in hindsight.



Classical DIP → Early deep learning → Modern SOTA



Horizontal position is chronological. The fork in 2018 is the point where the field split into two independent attacks on DnCNN's limitations.

Three limitations that each forced an era change

BOTTLENECK 1

Hand-designed priors, per-image cost

Ended: Classical DIP

TNRD observes that most state-of-the-art techniques of the period concentrated on utmost restoration quality with little consideration of computational efficiency. Compounding this: the prior's parameters and thresholds are chosen by the designer — the method cannot improve by being shown data.

Resolved by TNRD, then DnCNN.

BOTTLENECK 2

The clean-data requirement

Ended: supervised-only deep learning

Noise2Noise states the cost directly: obtaining clean training targets is often difficult or tedious — a noise-free photograph requires a long exposure. Noise2Void adds that in biomedical data, acquiring training targets, clean or noisy, is frequently not possible.

Resolved by the data branch: N2N → N2V → ZS-N2N.

BOTTLENECK 3

The limits of convolution

Ended: the CNN monopoly

Restormer names both weaknesses: convolution gives local connectivity and translation equivariance — efficiency and generalisation — but also a limited receptive field and inability to adapt to input content. MambaIR restates it as a dilemma between global receptive fields and efficient computation.

Three live answers: Restormer, NAFNet, MambaIR.

THE ARC IN ONE SENTENCE *The prior moves steadily out of human hands: written by a human (BM3D, WNNM) → structured by a human, fitted to data (TNRD) → learned from paired data (DnCNN) → learned without clean data (N2N), without pairs (N2V), without any data (ZS-N2N).*

PART III

The sources

15 distinct sources. 11 Category A at 78.6%, clearing the 70% floor with margin — and every Category A citation count verified against Google Scholar.



Eleven papers, 2007 → 2024

#	Short name	Lead author	Venue & year	Institution (lead)	Cited by	Era
A1	BM3D	Dabov et al.	IEEE TIP 16(8), 2007	Tampere Univ. of Tech.	11,976	I
A2	WNNM	Gu et al.	CVPR 2014, pp. 2862–2869	Hong Kong PolyU	2,960	I
A3	TNRD	Chen & Pock	IEEE TPAMI 39(6), 2017	Graz Univ. of Tech.	1,626	II
A4	DnCNN (anchor)	Zhang et al.	IEEE TIP 26(7), 2017	Harbin Inst. of Tech.	11,674	III
A5	Noise2Noise	Lehtinen et al.	ICML 2018 (PMLR 80)	NVIDIA · Aalto · MIT	3,064	IV-A
A6	Noise2Void	Krull et al.	CVPR 2019	MPI-CBG/PKS Dresden	2,075	IV-A
A7	Restormer	Zamir et al.	CVPR 2022, pp. 5728–5739	Inception Inst. of AI	6,181	IV-B
A8	NAFNet	Chen et al.	ECCV 2022, pp. 17–33	MEGVII Technology	2,318	IV-B
A9	ZS-N2N	Mansour & Heckel	CVPR 2023, pp. 14018–14027	TU Munich · Rice	266	IV-A
A10	SCUNet	Zhang et al.	Machine Intelligence Research 20(6), 2023	ETH Zürich	391	IV-B
A11	MambalR	Guo et al.	ECCV 2024, pp. 222–241	Tsinghua SIGS	1,208	IV-B
A12	Positive2Negative	Tong Li et al.	CVPR 2025, pp. 17924–17934	Beijing Institute of Technology	12	IV-A

RECENCY REQUIREMENT — SATISFIED BOTH WAYS

Six pre-2022 milestones (A1–A6) and five 2022-or-later SOTA papers (A7–A11). The guidelines require both halves; the roster carries both with margin. Citation counts are Google Scholar, checked 9 August 2026.

Three auxiliary sources — 21.4%, under the 30% ceiling

B1

Stanford CS231n — Convolutional Neural Networks for Visual Recognition

Recognised academic lecture series

B2

Google Research Blog — Night Sight: Seeing in the Dark on Pixel Phones

Industry-leader technical blog

B3

Gonzalez & Woods — Digital Image Processing, Image Restoration chapter

Textbook chapter

ALL THREE TYPES NAMED VERBATIM IN THE GUIDELINES

The guidelines list textbook chapters, Stanford/MIT lectures, and Google/Meta technical blogs as acceptable. Each Category B entry maps to one of those three named types exactly.

THE QUOTA, CHECKED			
A — peer-reviewed	12	80%	Pass
B — auxiliary	3	20%	Pass
Total	15	100%	Pass

WHY 15 AND NOT 10

At 10 sources the split would be 7/3, leaving no room to drop a paper if a venue is challenged. At 15 we can lose one Category A source and still sit at 11/3 — comfortably above the floor on both rules.

Category A is always rounded up: 70% is a floor, not a target.

What each auxiliary source is for

B1

Stanford CS231n

Recognised academic lecture series — named verbatim in the guidelines.

WHAT IT SUPPLIES

The mechanics our Category A papers assume rather than explain — convolution arithmetic, receptive-field growth, batch normalisation.

BRIDGE TO STAGE 2

DnCNN reports a 35×35 receptive field at depth 17 without deriving it. CS231n supplies the derivation — legible when we read real repository code.

B2

Google Research — Night Sight

Industry-leader technical blog — Google Research named verbatim in the guidelines.

WHAT IT SUPPLIES

The separation of photon shot noise from read noise, the SNR/exposure relationship, and the HDR+ burst capture / align / merge pipeline.

BRIDGE TO STAGE 2

The only source describing denoising as a shipped consumer product. It anchors Stage 2's deployment-reality requirement.

B3

Gonzalez & Woods

Textbook chapter on image restoration — an explicitly permitted type.

WHAT IT SUPPLIES

The mathematics of noise models — Gaussian, Rayleigh, Poisson, salt-and-pepper — and of mean, order-statistic and adaptive filters.

BRIDGE TO STAGE 2

The bottom rung of the timeline, below BM3D — the reference against which SCUNet's synthetic degradation pipeline can be read critically.

PART IV

Paper by paper

Metadata and credibility · technical specifications · evolutionary context. Eleven slides, one per Category A source, in the order the field produced them.

IV

A1 Image Denoising by Sparse 3-D Transform-Domain Collaborative Filtering

BM3D · Dabov, Foi, Katkovnik & Egiazarian

INSTITUTION

Institute of Signal Processing, Tampere University of Technology, Finland

VENUE & YEAR

IEEE Trans. Image Processing, 16(8), pp. 2080–2095, Aug 2007

CITATION COUNT

11,976 — Google Scholar, checked 9 Aug 2026

SUB-PROBLEM

Exploit non-local self-similarity to sparsify content enough that transform-domain shrinkage separates signal from noise.

DATASET(S)

Standard grayscale test images; a colour extension (C-BM3D). No training set — nothing is trained.

MODEL / ARCHITECTURE

Not a network. A four-stage algorithm: grouping by block matching → collaborative filtering via 3-D transform and shrinkage → inverse transform → aggregation of overlapping estimates by weighted averaging. Run twice; the second pass uses a collaborative Wiener filter with the first pass as an oracle.

VALUE THEN VS. NOW

Defined state of the art in 2007. Today it is the reference baseline — SCUNet still compares against BM3D-class methods sixteen years later.

RELATION TO THE OTHER PAPERS

It is the thing to beat. WNNM's abstract claims to outperform state-of-the-art algorithms such as BM3D by name. The entire learned-prior line exists because BM3D's hand-designed prior, however good, could not improve by being shown data.

A2 Weighted Nuclear Norm Minimization with Application to Image Denoising

WNNM · Gu, Zhang, Zuo & Feng

INSTITUTION

Hong Kong Polytechnic University · Harbin
Institute of Technology · Xidian University

VENUE & YEAR

CVPR 2014, pp. 2862–2869 · DOI
10.1109/CVPR.2014.366

CITATION COUNT

2,960 — Google Scholar, checked 9 Aug 2026

SUB-PROBLEM

Standard nuclear norm minimisation weights every singular value equally to preserve convexity — restricting flexibility where singular values carry distinct physical meaning.

DATASET(S)

Twenty widely used test images.

MODEL / ARCHITECTURE

An optimisation method, not a network. Weighted nuclear norm minimisation over matrices of non-local similar patches, analysed under different weighting conditions:

$$\|X\|_w = \sum w(i) \cdot \sigma(i)(X)$$

VALUE THEN VS. NOW

The strongest classical result in the roster and still a standard classical baseline — but a stepping stone, not a live method.

RELATION TO THE OTHER PAPERS

Shares BM3D's non-local self-similarity prior but reaches it via low-rank matrix approximation instead of transform-domain sparsity — showing the classical era's real content was the prior, not the machinery. Its cost motivates TNRD's efficiency argument.

A3 Trainable Nonlinear Reaction Diffusion

TNRD · Chen & Pock — the bridge paper

INSTITUTION

Institute for Computer Graphics and Vision, Graz
University of Technology, Austria

VENUE & YEAR

IEEE TPAMI, 39(6), pp. 1256–1272, 2017 (PDF
header reads 2016 acceptance; issue is 2017)

CITATION COUNT

1,626 — Google Scholar, checked 9 Aug 2026

SUB-PROBLEM

Restoration quality had been pursued with little
regard for computational efficiency, despite
efficiency being critical for real applications.

DATASET(S)

400 training images at 180×180 for denoising; Set5 and Set14 for super-resolution.

MODEL / ARCHITECTURE

A dynamic nonlinear reaction–diffusion model with time-dependent parameters. Critically, all parameters — linear filters and influence functions — are learned simultaneously from training data. Few diffusion steps; well suited to GPU parallelism.

VALUE THEN VS. NOW

Historically pivotal, practically superseded. Its importance is conceptual: it proved a shallow trainable feed-forward model could rival an iterative optimiser.

RELATION TO THE OTHER PAPERS

The hinge. The first paper in our list where the prior is learned while the computational structure stays hand-designed. DnCNN discards the diffusion scaffolding and keeps only the learning. Without TNRD, DnCNN is unmotivated.

A4 Beyond a Gaussian Denoiser: Residual Learning of Deep CNN for Image Denoising

DnCNN · Zhang, Zuo, Chen, Meng & Zhang — ANCHOR PAPER

INSTITUTION

Harbin Inst. of Tech. · Hong Kong PolyU · Graz Univ. of Tech. · Xi'an Jiaotong University

VENUE & YEAR

IEEE Trans. Image Processing, 26(7), pp. 3142–3155, 2017 · DOI 10.1109/TIP.2017.2662206

CITATION COUNT

11,674 — Google Scholar, checked 9 Aug 2026

SUB-PROBLEM

Existing discriminative denoisers trained a separate model per noise level. DnCNN targets blind Gaussian denoising with one model.

DATASET(S)

BSD68, Set12 (denoising); Set5, Set14, Urban100 (super-resolution); Classic5, LIVE1 (JPEG deblocking).

MODEL / ARCHITECTURE

Deep feed-forward CNN combining residual learning with batch normalisation, predicting the noise rather than the image. Depth 17, receptive field 35×35.

$$\hat{x} = y - R(y)$$

VALUE THEN VS. NOW

Established residual learning + BN as the default denoising recipe and remains a universal baseline — every modern paper here still reports against it. Superseded as a deployed method.

RELATION TO THE OTHER PAPERS

The hinge of the whole review. It resolves TNRD's residual hand-design; it is the supervised method whose data requirement N2N → N2V → ZS-N2N dismantle; and it is the convolutional method whose receptive-field limits Restormer and MambaIR are written against.

A5 Noise2Noise: Learning Image Restoration without Clean Data

N2N · Lehtinen, Munkberg, Hasselgren, Laine, Karras, Aittala & Aila

INSTITUTION

NVIDIA · Aalto University · MIT CSAIL

VENUE & YEAR

ICML 2018 — Proceedings of the 35th Intl. Conf. on Machine Learning, PMLR 80

CITATION COUNT

3,064 — Google Scholar, checked 9 Aug 2026

SUB-PROBLEM

Clean training targets are difficult or tedious to obtain — the paper's own example: a noise-free photograph requires a long exposure.

DATASET(S)

Kodak, Set14, ImageNet-derived training data; also Monte Carlo renderings and undersampled MRI scans.

MODEL / ARCHITECTURE

A training paradigm, not an architecture. The empirical risk minimised is unchanged from supervised learning the insight is that the target y_i may itself be a noisy observation.

VALUE THEN VS. NOW

Conceptually foundational and still actively built upon — ZS-N2N, in this same roster, states it is motivated by Noise2Noise.

RELATION TO THE OTHER PAPERS

The first break from DnCNN's supervision. Strictly weaker in data requirements than DnCNN, strictly stronger than Noise2Void. Still needs pairs of independent noisy observations of the same scene — the requirement N2V removes.

A6 Noise2Void — Learning Denoising from Single Noisy Images

N2V · Krull, Buchholz & Jug

INSTITUTION

MPI-CBG / PKS (CSBD), Dresden, Germany

VENUE & YEAR

CVPR 2019

CITATION COUNT

2,075 — Google Scholar, checked 9 Aug 2026

SUB-PROBLEM

Even noisy pairs are usually unavailable. In biomedical imaging, acquiring training targets — clean or noisy — is frequently not possible.

DATASET(S)

BSD68 among the evaluation sets, plus biomedical imagery.

MODEL / ARCHITECTURE

A training scheme built on a blind-spot network: the receptive field is constructed to exclude the centre pixel, so the network cannot solve the task by copying its input and must predict each pixel from its context.

VALUE THEN VS. NOW

Still in active use in microscopy and biomedical imaging — precisely the domain it was designed for. The paper is candid that performance drops moderately, given strictly less information available.

RELATION TO THE OTHER PAPERS

Removes the one requirement N2N retained. Its own abstract positions it as taking the N2N idea one step further — the N2N → N2V → ZS-N2N chain is evidenced by the papers themselves, not imposed by us.

A7 Restormer: Efficient Transformer for High-Resolution Image Restoration

Zamir, Arora, S. Khan, Hayat, F. S. Khan & Yang

INSTITUTION

Inception Inst. of AI · MBZUAI · Monash · UC
Merced · Yonsei · Google Research

VENUE & YEAR

CVPR 2022, pp. 5728–5739

CITATION COUNT

6,181 — Google Scholar, checked 9 Aug 2026

SUB-PROBLEM

Transformer complexity grows quadratically with spatial resolution, making it infeasible for most high-resolution restoration tasks.

DATASET(S)

SIDD, DND (real denoising); Urban100, Kodak, Set12, McMaster, BSD68 (Gaussian); GoPro, HIDE (deblurring); Rain100 (deraining). Trained only on SIDD, tested directly on DND.

MODEL / ARCHITECTURE

Encoder–decoder Transformer from two redesigned blocks: MDTA (multi-Dconv head transposed attention), aggregating local/non-local interactions efficiently; and GDFN (gated-Dconv feed-forward network), gating so only useful information propagates. No local windows — preserves distant context.

VALUE THEN VS. NOW

Current and heavily used. The canonical Transformer endpoint of the arc the guidelines require.

RELATION TO THE OTHER PAPERS

Answers DnCNN's receptive-field limitation directly and by name. NAFNet is its direct methodological rival; SCUNet borrows the transformer half of its hybrid from this line.

A8 Simple Baselines for Image Restoration

NAFNet · Chen, Chu, Zhang & Sun — the dissenting voice

INSTITUTION

MEGVII Technology, Beijing, China

VENUE & YEAR

ECCV 2022, LNCS 13667, pp. 17–33, Springer

CITATION COUNT

2,318 — Google Scholar, checked 9 Aug 2026

SUB-PROBLEM

Rising system complexity in SOTA methods hinders convenient analysis and comparison — decomposed into inter-block and intra-block complexity, both attacked.

DATASET(S)

SIDD (denoising) and GoPro (deblurring), used throughout the ablations.

MODEL / ARCHITECTURE

A Nonlinear Activation Free Network derived by progressively simplifying a plain UNet baseline. Central claim: nonlinear activations (Sigmoid, ReLU, GELU, Softmax) are not necessary and can be replaced by multiplication (SimpleGate) or removed, alongside simplified channel attention.

REPORTED RESULTS

40.30 dB PSNR on SIDD, exceeding previous SOTA by 0.28 dB with less than half the compute. 33.69 dB on GoPro, exceeding previous SOTA by 0.38 dB with only 8.4% of its cost.

RELATION TO THE OTHER PAPERS

The roster's counter-argument. Where Restormer argues for more expressive blocks, NAFNet argues much of that complexity is unnecessary — and backs it with numbers. Including both stops the narrative becoming a naive newer-is-better story.

A9 Zero-Shot Noise2Noise: Efficient Image Denoising without any Data

ZS-N2N · Mansour & Heckel

INSTITUTION

Technical University of Munich · MCML · Rice University

VENUE & YEAR

CVPR 2023, pp. 14018–14027 · DOI 10.1109/CVPR52729.2023.01347

CITATION COUNT

266 — Google Scholar, checked 9 Aug 2026.
Lower count reflects recency, not weakness.

SUB-PROBLEM

Existing dataset-free methods are either computationally expensive, require a noise model, or give inadequate image quality.

DATASET(S)

Artificial noise, real-world camera noise and microscope noise — Kodak, SIDD, PolyU, McMaster among the benchmarks.

MODEL / ARCHITECTURE

A simple 2-layer network trained per image. No training data, no knowledge of the noise distribution. Its only assumption is pixel-wise independence of the noise.

VALUE THEN VS. NOW

Current, and the most immediately reproducible source in the roster — the paper points to a Colab notebook with its implementation and hyperparameters.

RELATION TO THE OTHER PAPERS

The terminus of the data branch: DnCNN needs clean pairs → N2N needs noisy pairs → N2V needs a noisy dataset → ZS-N2N needs nothing but the image in front of it. The paper names Noise2Noise as its motivation, closing the loop inside our own list.

A10 Practical Blind Image Denoising via Swin-Conv-UNet and Data Synthesis

SCUNet · Zhang, Li, Liang, Cao, Zhang, Tang, Fan, Timofte & Van Gool

INSTITUTION

Computer Vision Lab, ETH Zürich · University of Würzburg · KU Leuven

VENUE & YEAR

Machine Intelligence Research, 20(6), pp. 822–836, 2023 · DOI 10.1007/s11633-023-1466-0

CITATION COUNT

391 — Google Scholar, checked 9 Aug 2026

SUB-PROBLEM

Existing methods rely on simple noise assumptions; a general-purpose blind denoiser for real images remains unsolved.

DATASET(S)

Urban100, Set12, Kodak, DND, CBSD68, BSD68, McMaster, SIDD, Waterloo, DIV2K.

MODEL / ARCHITECTURE

A swin-conv block combining the local modelling of a residual convolutional layer with the non-local modelling of a swin transformer block, plugged into an image-to-image translation UNet. Second contribution: a practical degradation model spanning Gaussian, Poisson, speckle, JPEG compression and processed camera sensor noise, with random-shuffle and double-degradation strategies.

VALUE THEN VS. NOW

The architecture reaches SOTA and the degradation model significantly improves practicability, per the paper's own findings.

RELATION TO THE OTHER PAPERS

The synthesis paper. It fuses the convolutional line (DnCNN) with the transformer line (Restormer/SwinIR-style blocks), and is the only source that treats the noise model itself — the assumption in DnCNN's opening equation — as the thing to fix.

A11 MambaIR: A Simple Baseline for Image Restoration with State-Space Model

Guo, Li, Dai, Ouyang, Ren & Xia — the current frontier

INSTITUTION

Tsinghua SIGS · Shenzhen University · ByteDance Inc. · Aitist.ai · Peng Cheng Lab

VENUE & YEAR

ECCV 2024, pp. 222–241 (arXiv:2402.15648v3, 15 Oct 2024)

CITATION COUNT

1,208 — Google Scholar, checked 9 Aug 2026

SUB-PROBLEM

The dilemma between global receptive fields and efficient computation — plus two Mamba failures in low-level vision: local pixel forgetting and channel redundancy.

DATASET(S)

Urban100, Manga109, Set5, Set14, DIV2K, Flickr2K (super-resolution); SIDD for training, SIDD test + DND for real denoising.

MODEL / ARCHITECTURE

A Residual State Space Block (RSSB) adding local enhancement and channel attention to the vanilla Mamba selective state space model — giving long-range dependency modelling at linear complexity rather than quadratic.

REPORTED RESULT

Outperforms SwinIR by up to 0.45 dB on image super-resolution at similar computational cost, but with a global receptive field.

RELATION TO THE OTHER PAPERS

Resolves the same dilemma Restormer attacked, but by replacing attention altogether rather than making it cheaper. Together with NAFNet, shows the field has three live answers to bottleneck 3.

A12 Positive2Negative: Breaking the Information-Lossy Barrier in Self-Supervised Single Image Denoising

Li, Wang, Xu, Zhu, Lu & Huang

INSTITUTION

School of Computer Science and Technology,
Huazhong University of Science and Technology,
China

VENUE & YEAR

IEEE/CVF Conference on Computer Vision and
Pattern Recognition (CVPR), 17924–17934, 2025

CITATION COUNT

12 — Google Scholar, checked 24 Aug 2026

SUB-PROBLEM

Existing self-supervised denoising methods rely on information-lossy operations such as masking and down sampling. These operations can remove useful high-frequency details from the input image.

DATASET(S)

SIDD and standard synthetic- and real-noise benchmarks. No clean training targets or paired noisy images are required during self-supervised training.

MODEL / ARCHITECTURE

Not a completely new backbone. A self-supervised training paradigm based on Re-noised Data Construction and Denoised Consistency Supervision. Predicted noises used to construct positive and negative full-resolution noisy variants, whose denoised outputs are forced to agree.

REPORTED RESULT

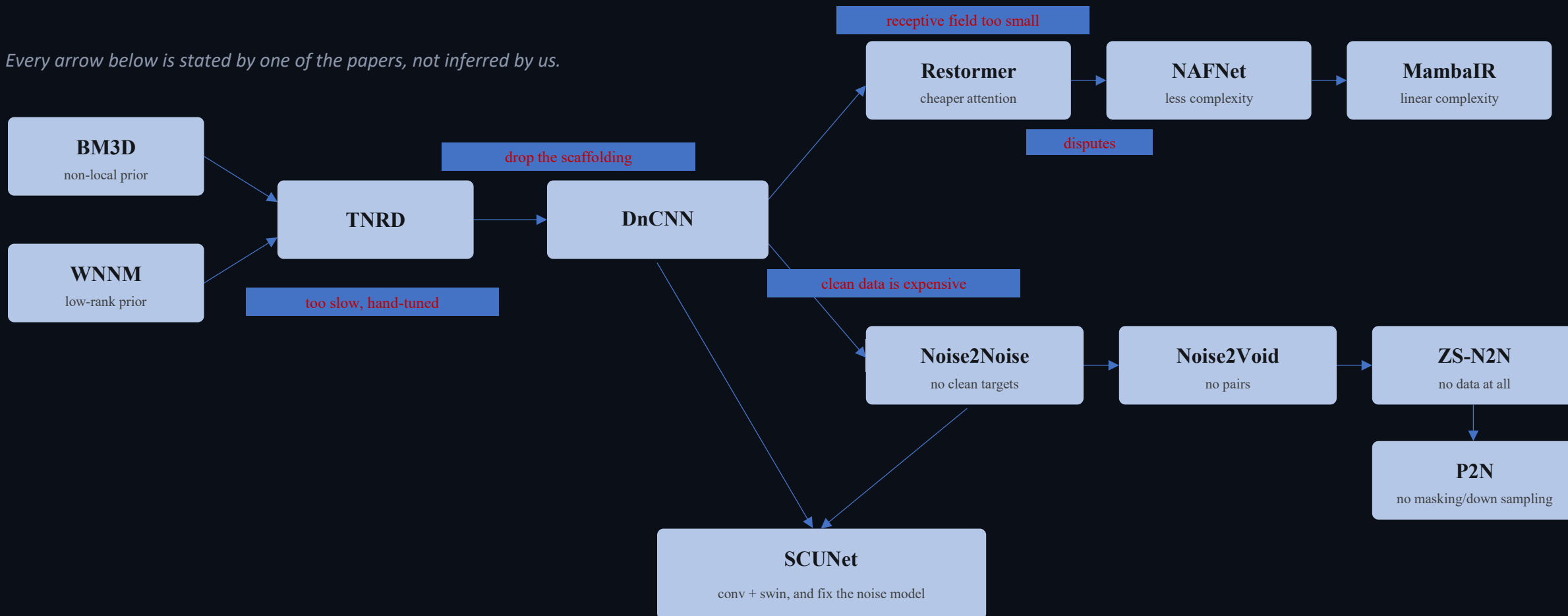
A major 2025 advance in self-supervised single-image denoising. P2N reports state-of-the-art performance with improved training efficiency while avoiding the information loss caused by masking and down sampling.

RELATION TO THE OTHER PAPERS

P2N continues the Noise2Noise → Noise2Void → ZS-N2N self-supervised branch. It removes the need for clean targets and noisy pairs, while also avoiding blind-spot masking and down sampling. Its main contribution is to preserve the complete image information during training.

How the eleven papers depend on each other

Every arrow below is stated by one of the papers, not inferred by us.



Solid arrows mean "builds on"; the dashed arrow means "disputes" — a disagreement, not a dependency. SCUNet fuses both lines and closes the arc DnCNN opened, by attacking the noise assumption itself.

THE ASK

The research direction we are proposing

Image denoising is the cleanest available case study in how computer vision replaced hand-designed priors with learned ones. Its history divides into four eras separated by three identifiable bottlenecks — manual prior design, the clean-data requirement, and the receptive-field limits of convolution.

Each bottleneck was named by the paper that broke it, so the arc is evidenced from primary sources rather than imposed in hindsight. And because the field currently holds three competing answers to the third bottleneck, the story ends in a genuine open question — not a winner.

References

CATEGORY A — 12

[A1] K. Dabov, A. Foi, V. Katkovnik, K. Egiazarian. Image Denoising by Sparse 3-D Transform-Domain Collaborative Filtering. IEEE TIP, 16(8):2080–2095, 2007.

[A2] S. Gu, L. Zhang, W. Zuo, X. Feng. Weighted Nuclear Norm Minimization with Application to Image Denoising. CVPR, pp. 2862–2869, 2014.

[A3] Y. Chen, T. Pock. Trainable Nonlinear Reaction Diffusion. IEEE TPAMI, 39(6):1256–1272, 2017.

[A4] K. Zhang, W. Zuo, Y. Chen, D. Meng, L. Zhang. Beyond a Gaussian Denoiser. IEEE TIP, 26(7):3142–3155, 2017.

[A5] J. Lehtinen et al. Noise2Noise: Learning Image Restoration without Clean Data. ICML, PMLR 80, 2018.

[A6] A. Krull, T.-O. Buchholz, F. Jug. Noise2Void. CVPR, 2019.

[A7] S. W. Zamir et al. Restormer: Efficient Transformer for High-Resolution Image Restoration. CVPR, pp. 5728–5739, 2022.

[A8] L. Chen, X. Chu, X. Zhang, J. Sun. Simple Baselines for Image Restoration. ECCV, LNCS 13667, pp. 17–33, 2022.

[A9] Y. Mansour, R. Heckel. Zero-Shot Noise2Noise. CVPR, pp. 14018–14027, 2023.

[A10] K. Zhang et al. Practical Blind Image Denoising via Swin-Conv-UNet and Data Synthesis. MIR, 20(6):822–836, 2023.

[A11] H. Guo et al. MambaIR: A Simple Baseline for Image Restoration with State-Space Model. ECCV, pp. 222–241, 2024.

[A12] T. Li et al. Positive2Negative: Breaking the Information-Lossy Barrier in Self-Supervised Single Image Denoising. CVPR, pp. 17924–17934, 2025.

CATEGORY B — 3

[B1] Stanford University. CS231n: Convolutional Neural Networks for Visual Recognition. Lecture series.

[B2] Google Research. Night Sight: Seeing in the Dark on Pixel Phones. Google Research Blog.

[B3] R. C. Gonzalez, R. E. Woods. Digital Image Processing — Image Restoration chapter.

Note: Venues for A1, A3, A5 and A10 are printed in the source PDFs themselves. Venues for A2, A4, A6, A7, A8, A9 and A11 were verified against publisher, CVF Open Access or institutional proceedings records, because those PDFs are preprint versions that do not carry their venue.